Stage3G: executed bounded development experiment

Fixed three K30 SiteID pools
Source training: 2016–2021
No protected group or
2024–2025 performance

Source selection / cross-fitting
Base fitting + descriptors end 2020
Pseudo-deployments: Jan/Jul 2021, 14 origins

Five group-isolated outer folds
Base fit: 12 groups; risk train: 6
Threshold calibration: 6; evaluation: 6

Inner three-fold group CV on base-fit groups only
Alpha/lambda/eta selection and descriptor scaling
Held-out outer groups excluded upstream

Source base prediction is sample-out
Only each role's permitted initial B days
Persistence + residual Ridge + legal local bias

Risk-training role: six groups
Fixed logistic regression, three inputs
Imputation + standardization fitted here

Separate threshold-calibration role
Fixed 10% / 20% expected review budgets
Truth-independent hash handling of score ties

Held-out risk scores: six evaluation groups
Independent from risk fit and threshold calibration
Repeat across the five outer folds

Frozen target risk model: fit on source base-OOF rows; operating thresholds from honest outer risk scores
Source-to-target calibration transfer remains a limitation; target deployment labels never flow upstream

Target initial B calendar days
[T−B, T−1], B=30 / 90 / 180
Scale, descriptors, local training,
validation and bias
No observations before budget left boundary

Final source fit: 2016–2021 only
Uniform / similarity source weights
Scale, weights, alpha, lambda and bias freeze at T
B30 main-pool fixed weight-shuffle ablation

60 deployment origins, h=1 / h=3
Update only legally arrived lag inputs
Risk: initial distance + recent missingness/change
No retraining or threshold recalibration

Saved point predictions + pre-forecast risk scores
Fixed review flags; prediction remains unchanged
Raw predictions primary; uniform nonnegative sensitivity
h=3 predicts origin+3 daily volume

Evaluation only: ten fixed development SiteIDs × five USED 2022–2023 windows × all budgets/horizons/pools
Matched-date MAE/RMSE/negative transfer and history comparisons; risk within-site/window and between-condition statistics
5000 paired synchronous calendar-block + SiteID resamples; four primary contrasts; no independent physical-facility or real-time certification